

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2022 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

4049/01

Paper 1

29 August 2022
2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Calculator Model

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Total Marks

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}.$

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}bc \sin A$$

1 (a) Factorise $y^3 - 125$.

[1]

(b) Hence write 7875 as the product of two integers.

[2]

- 2 The triangle XYZ is such that its area is $\frac{1}{4}(28 + \sqrt{2}) \text{ cm}^2$, the length of XY is $(5 + \sqrt{2}) \text{ cm}$ and angle XYZ is 45° .

Find, **without using a calculator**, the length YZ , in cm, in the form $a\sqrt{2} + b$, where a and b are integers. [4]

- 3 Express $\frac{6x^3 - 29x^2 + 4x - 23}{2x^2 - 9x - 5}$ in partial fractions. [6]

- 4 **(a)** Given that $ax^2 + 7x + b$ is not greater than 8, what conditions must apply to the constants a and b ? [4]

- (b)** Give an example of possible values for a and b which satisfy the conditions in **part (a)**. [2]

5 The equation of a curve is $y = \frac{\sin 3x}{\cos x - 2}$.

The point P lies on the curve and has an x -coordinate of $\frac{\pi}{3}$.

(a) Find the equation of the tangent to the curve at P . [5]

The tangent to the curve at P meets the y -axis at Q .

(b) Find the exact values of the coordinates of Q . [1]

- 6 [The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$ and the surface area of a sphere of radius r is $4\pi r^2$.]

Air is pumped, at a constant rate of $42 \text{ cm}^3/\text{s}$, into a spherical balloon of radius $r \text{ cm}$ such that the surface area of the balloon, $A \text{ cm}^2$, and volume of air in the balloon, $V \text{ cm}^3$, are increasing.

Assuming that the balloon retains its shape, calculate the rate of change of the surface area of the balloon when its radius is 15 cm . [7]

7 (a) Solve the equation $\log_3 p + 4 = \log_{27} p$. [4]

(b) It is given that $\ln q + \ln r + \ln 1 = \ln(q + r + 1)$.

(i) Express q in terms of r . [2]

(ii) State the range of values of q and explain clearly why $r > 1$. [2]

8 (a) Prove that $\frac{\sin x}{\cot x + \sin x - \operatorname{cosec} x} = \sec x + 1$. [4]

(b) Hence solve the equation $\frac{\sin 2x}{\cot 2x + \sin 2x - \operatorname{cosec} 2x} = \tan^2 2x$ for $0^\circ < x < 90^\circ$. [4]

- 9 The function f is defined, for all values of x , by $f(x) = x^4 - 4x^3 + ax^2 + bx + 8$, where a and b are integers.

The graph of $y = f(x)$ has a stationary point at $(1, -1)$.

- (a) Find the value of a and of b .

[5]

- (b) Explain, with working, whether $f'(x)$ is an increasing or a decreasing function.

[3]

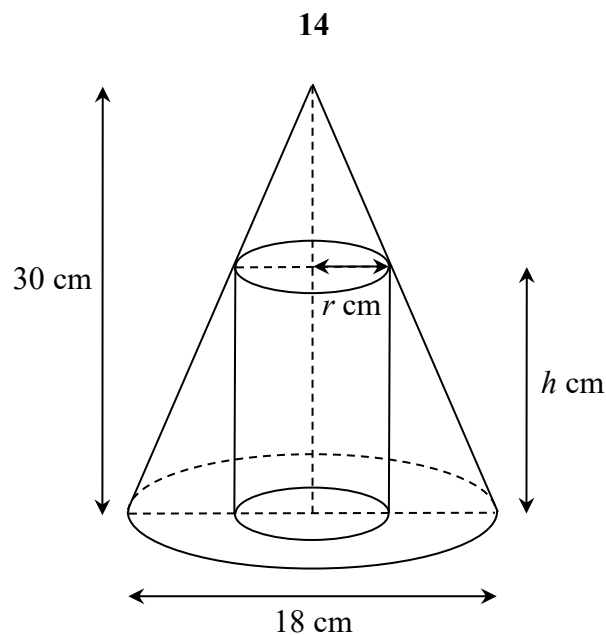
- 10 The acute angles A and B are such that $\tan(A+B) = 7$ and $\tan B = 3$.

Without using a calculator, find the exact value of

(a) $\cos A$, [5]

(b) $\sin 2B$. [3]

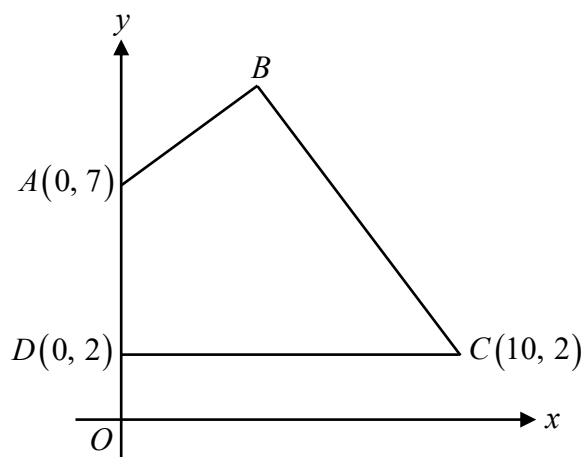
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The diagram shows a candle in the shape of a right circular cone of base diameter 18 cm and height 30 cm. A solid cylinder of radius r cm and height h cm is to be carved out of this cone.

- (a) Show that the volume, $V \text{ cm}^3$, of the cylinder is given by $V = 30\pi r^2 - \frac{10}{3}\pi r^3$. [3]

- (b) Given that r can vary, find the value of r for which there is a stationary value of V and calculate this value of V . [5]



The diagram shows a kite $ABCD$ in which $AB = AD$ and $BC = CD$.

The coordinates of A , C and D are $(0, 7)$, $(10, 2)$ and $(0, 2)$ respectively.

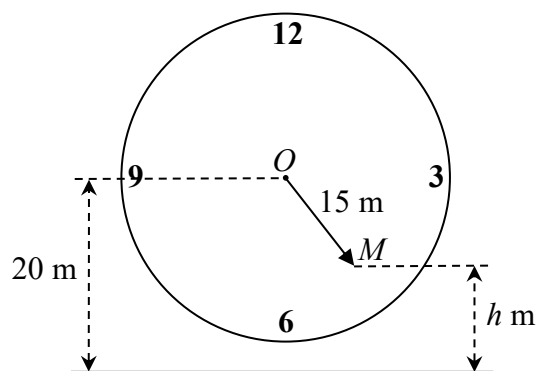
(a) Find the equation of BD .

[3]

(b) Hence show that the coordinates of B are $(4, 10)$. [4]

(c) Find the coordinates of the point equidistant from A , B and D . [2]

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A park has a clock set on a vertical wall.

The distance from the centre of the clock, O , to the tip, M , of the minute hand is 15 m.

The height, h m, of M above ground level is given by $h = a \cos kt + b$, where t is the time in minutes past the hour.

O is 20 m above the ground.

(a) Find the value of a and of b .

[2]

(b) Show that the value of k is $\frac{\pi}{30}$ radians per minute.

[2]

(c) Sketch the graph of h for $0 \leq t \leq 45$.

[2]

(d) Find the duration in each hour such that $h < 10$.

[3]

END OF PAPER

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